



# 实变函数八大定理总结

# 实变函数八大定理

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Ù/ Ž2026tÒGêfžØýp ;Ó,, ° ...¹:;Ñp žØýp  
Í•„k\*š 'è f!•Bf ŸÃÆá Ø™ fì„Á Ç s¿v-'  
è ùŽ,Ñ ¾¼åÁ  
Ùk\*š „Á Ç t ÷ ( w v(º{ý ã„í €•²ã  
† ĭ å¿ êñ ` ö÷ ×?ö•  
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# 1. W+(Egorov)š

## 1.1. š ...<sup>1</sup>

$\frac{3}{4} E \in R^n$  :ikÆ  $m(E) < +1$  ff  $n g / E$  „ikýp  $f / E \in N$   
 $P$  „ikýp á  $f$   $n! \int_{\mathbb{R}^n} E \cdot \hat{u} > 0$  X(ikpÆ  $E$  „  $E \in \mathcal{A}$   
<sup>3</sup>

$$m(E \cap E_n) < \epsilon;$$

ff  $n g (E_n) \in \mathcal{O}(\epsilon)$

## 1.2. Á ĩ

- $6[\cdot; \cdot; \cdot] \in \mathbb{R}^N$   $S_1 \cdot T_1 \cdot E_{n;k} = \sum_{n=N}^{\infty} \int_{\mathbb{R}^n} |f_n(x) - f(x)| < \frac{1}{k}$   $f_n \in \mathcal{A}$
- $K_1 \cdot \mathcal{P} \cdot \mathcal{U}^* \cdot k \in X(N)$   $k \cdot m \cdot E_n \cdot T_1 \cdot E_{n;k} < \frac{1}{2^k}$
- „  $\mathcal{O}(\epsilon) \in \mathcal{A}$   $\epsilon = \sum_{k=1}^{\infty} \sum_{n=N_k}^{\infty} E_{n;k} \in \mathcal{K}_1 \cdot \mathcal{A} \cdot \mathcal{O}(\epsilon)$

## 1.3. Á

$\hat{u} \in \mathcal{C}^k$  š

$$E_{n;k} = \sum_{n=N}^{\infty} \int_{\mathbb{R}^n} |f_n(x) - f(x)| < \frac{1}{k};$$

$f_n(x) \cdot f(x) \in \mathcal{L}^1(\mathbb{R}^n)$   $\mathcal{A} \cdot \mathcal{S} \cdot \hat{u} \in X(N) \cdot n \cdot N \cdot \sum_{n=N}^{\infty} \int_{\mathbb{R}^n} E_{n;k} \in \mathcal{S}$

$$\sum_{k=1}^{\infty} \sum_{n=N}^{\infty} \int_{\mathbb{R}^n} |f_n(x) - f(x)| = \sum_{k=1}^{\infty} \sum_{n=N}^{\infty} E_{n;k} :$$

$1 \cdot f_n \in \mathcal{A}$

$$\sum_{k=1}^{\infty} \sum_{n=N}^{\infty} \int_{\mathbb{R}^n} E_{n;k} = 0;$$

$\mathcal{S} \cdot \mathcal{U}^* \cdot \mathcal{U} \cdot k$

$$\sum_{N=1}^{\infty} \sum_{n=N}^{\infty} \int_{\mathbb{R}^n} E_{n;k} = 0:$$

$9n \cdot K_1 \cdot \mathcal{P} \cdot \mathcal{U}^* \cdot k \in X(\mathcal{C}^k N)$

$$\sum_{n=N_k}^{\infty} \int_{\mathbb{R}^n} E_{n;k} < \frac{1}{2^k} :$$

„ $\mathcal{A}$

$$E'' = \bigvee_{k=1}^1 \bigvee_{n=N_k}^1 E_{n;k};$$

$$E \wedge E'' = \bigwedge_{k=1}^1 E \wedge \bigvee_{n=N_k}^1 E_{n;k};$$

$1K_{\mathcal{I},\mathcal{I}'}^{\mathcal{I},\mathcal{I}'}$

$$\begin{aligned} m(E \wedge E'') &= \bigwedge_{k=1}^1 m \bigvee_{n=N_k}^1 E_{n;k} \\ &< \bigwedge_{k=1}^1 \frac{1}{2^k} = 0. \end{aligned}$$

Á ff<sub>n</sub>g (E''<sub>T</sub><sup>1</sup><sub>n=N<sub>k<sub>0</sub></sub> E<sub>n;k<sub>0</sub></sub> sS n N<sub>k<sub>0</sub></sub> ö</sub>

$$0 \bullet \frac{1}{k_0} < \hat{u}$$

$$|f_n(x) - f(x)| < \frac{1}{k_0} < \delta;$$

1 ö6[„šI ff<sub>n</sub>g (E''<sub>T</sub><sup>1</sup><sub>n=N<sub>k<sub>0</sub></sub> E<sub>n;k<sub>0</sub></sub> sS n N<sub>k<sub>0</sub></sub> ö</sub>





$$\int_n (F_n - \sum_{k=1}^n F_k) / \epsilon$$

$$m(E_n^0) = \sum_{n=1}^{\infty} \frac{1}{2^{n+1}} = \frac{1}{2}.$$

ad

$$m(E_n) = m(E_n \cap E^0) + m(E_n \cap E^0)^c < \frac{1}{2} + \frac{1}{2} = 1.$$

$$\int_n (F_n - \sum_{k=1}^n F_k) / \epsilon = \int_n (F_n - \sum_{k=1}^n F_k) / \epsilon$$



## 4 ±ô(Levi)š

### 4.1 š ...<sup>1</sup>

$$\frac{3}{4} f f_n g / i K \mathcal{A} E E R \quad n \text{ „}^{\wedge} i K \acute{y} p \acute{a}^3$$

$$f_1(x) f_2(x) \quad f_n(x) :: (U \check{z});$$

$$f_n(x)! \overset{a:e}{f}(x)(n!1) \check{z} E$$

$$\lim_{n!1} \int_E f_n(x) dx = \int_E f(x) dx:$$

$$\grave{e} \text{ I } \$ \ddot{u} \grave{a} \ddot{o}: +1$$

### 4.2 Á ĩ

$$\bullet \bigcup_R' \text{ — } L 1 f \quad n \quad f_{n+1} \quad f \ddot{u} \bullet f \quad R_E f_n g U \check{z} E \lim_{n!1} R_E f_n$$

$$\bullet \text{ „} \mathcal{A} \text{ — } L \grave{u} \hat{u}^{\wedge} \epsilon U \acute{y} p' \quad f \ddot{O} \quad 2 (0; 1) \grave{a} E \quad n = f x \quad 2 E j$$

$$f_n(x) (1 \quad )'(x) g \quad E \quad n \% E$$

$$\bullet T \ddot{E} \text{ I } 1 K'_i \text{ } \mathfrak{P} \acute{u}' \text{ — } \lim_{n!1} R_E f_n (1 \quad ) \quad R_E' \grave{a} \quad ! 0 \text{ — } \acute{u}$$

$$\text{ I } \quad \grave{E} \text{ I } \ddot{E}$$

### 4.3 Á

$$4.3.1 \quad \acute{A} \lim_{n!1} R_E f_n \quad R_E f$$

$$\grave{a} f_n(x) f_{n+1}(x) f(x) \grave{u} \grave{a} N @ \quad x \quad 2 E \quad \ddot{E} E \acute{u} \bullet \quad R_E f_n(x) dx \quad / U$$

$$\check{z} \text{ „} \grave{a} d \bullet P (P \grave{a} \grave{u}) X ($$

$$\lim_{n!1} \int_E f_n(x) dx = \int_E f(x) dx:$$

$$4.3.2 \quad \acute{A} \lim_{n!1} R_E f_n \quad R_E f$$

$$\hat{u} \ddot{O}^{\wedge} \epsilon U \acute{y} p'(x) \acute{a}^3'(x) f(x) (x \quad 2 E) \hat{u} \ddot{O} \quad 2 (0; 1) \acute{s} \text{ I } \mathcal{A}$$

$$E_n = f x \quad 2 E j f_n(x) (1 \quad )'(x) g:$$

$$\grave{a} f_n(x) U \check{z} f \quad n(x)! \overset{a:e}{f}(x) E E \quad n \quad E_{n+1} \quad S_{n=1}^1 E_n = E (d \ddot{o} K \mathcal{A}$$

$$)$$

$$1 K'_i \text{ „ } \mathfrak{P} \acute{u}'$$

$$\lim_{n!1} m(E_n) = m(E):$$

$$\int_{E_n} f_n(x) dx \rightarrow \int_E f(x) dx \quad (1)$$

„U“

$$\lim_{n \rightarrow \infty} \int_{E_n} f_n(x) dx = \int_E f(x) dx$$

„0“

$$\lim_{n \rightarrow \infty} \int_{E_n} f_n(x) dx = \int_E f(x) dx$$

„f(x)“

—

$$\lim_{n \rightarrow \infty} \int_{E_n} f_n(x) dx = \int_E f(x) dx$$

„ü“

$$\lim_{n \rightarrow \infty} \int_{E_n} f_n(x) dx = \int_E f(x) dx$$

## 5 Ò <§66[š

### 5.1 š ...<sup>1</sup>

$\frac{3}{4} \int_E f_n(x) dx \leq \int_E f(x) dx$  for all  $E \in \mathcal{E}$ . Then  $\lim_{n \rightarrow \infty} \int_E f_n(x) dx = \int_E f(x) dx$ .

— If  $f_n(x) \rightarrow f(x)$  a.e. and  $\int_E f_n(x) dx \rightarrow \int_E f(x) dx$  for all  $E \in \mathcal{E}$ , then  $f_n \rightarrow f$  in  $L^1(E)$ .

$$\lim_{n \rightarrow \infty} \int_E f_n(x) dx = \int_E f(x) dx:$$

### 5.2 Á ĩ

- If  $f_n \rightarrow f$  a.e. and  $\int_E f_n(x) dx \rightarrow \int_E f(x) dx$  for all  $E \in \mathcal{E}$ , then  $f_n \rightarrow f$  in  $L^1(E)$ .
- (Fatou's lemma) If  $f_n \geq 0$  a.e. and  $\int_E f_n(x) dx \rightarrow \int_E f(x) dx$  for all  $E \in \mathcal{E}$ , then  $f_n \rightarrow f$  in  $L^1(E)$ .
- If  $f_n \rightarrow f$  a.e. and  $\int_E f_n(x) dx \rightarrow \int_E f(x) dx$  for all  $E \in \mathcal{E}$ , then  $f_n \rightarrow f$  in  $L^1(E)$ .

### 5.3 Á

If  $f_n \rightarrow f$  a.e. and  $\int_E f_n(x) dx \rightarrow \int_E f(x) dx$  for all  $E \in \mathcal{E}$ , then  $f_n \rightarrow f$  in  $L^1(E)$ .

à  $F(x) = f(x)$  a.e. 1Fatou

$$\int_E (F(x) - f_n(x)) dx = \lim_{n \rightarrow \infty} \int_E (F(x) - f_n(x)) dx;$$

U —

$$\int_E F(x) dx = \int_E f(x) dx = \lim_{n \rightarrow \infty} \int_E F(x) dx = \lim_{n \rightarrow \infty} \int_E f_n(x) dx;$$

€ —

$$\lim_{n \rightarrow \infty} \int_E f_n(x) dx = \int_E f(x) dx: \quad (1)$$

à  $F(x) = f(x)$  a.e. 1Fatou

$$\int_E (F(x) + f_n(x)) dx = \lim_{n \rightarrow \infty} \int_E (F(x) + f_n(x)) dx;$$

U —

$$\int_E F(x) dx + \int_E f_n(x) dx = \lim_{n \rightarrow \infty} \int_E F(x) dx + \lim_{n \rightarrow \infty} \int_E f_n(x) dx;$$

€ —

$$\int_E f(x) dx = \lim_{n \rightarrow \infty} \int_E f_n(x) dx: \quad (2)$$

Ó (1)(2)

$$\lim_{n \rightarrow \infty} \int_E f_n dx = \int_E f dx$$

$$\lim_{n \rightarrow \infty} \int_E f_n dx;$$

E •Pøls

$$\lim_{n \rightarrow \infty} \int_E f_n(x) dx = \int_E f(x) dx:$$

## 6 Ōp(Fatou)

### 6.1 Š ...<sup>1</sup>

$$\frac{3}{4} \text{ ff } f_n(x) \text{ g } /iK\text{Æ E } R \quad n \text{ „} \wedge \text{ iKýp}$$

$$\lim_{n \rightarrow \infty} \int_E f_n(x) dx = \lim_{n \rightarrow \infty} \int_E f_n(x) dx:$$

### 6.2 Á ĭ

- „ U ä g  $g_k(x) = \inf_{n \geq k} f_n(x)$  g  $g_k(x) \wedge U \checkmark \lim_{k \rightarrow \infty} g_k(x) =$   
 $\lim_{n \rightarrow \infty} f_n(x)$
- l 1 g  $\int_k f_n(n-k) — \int_E g_k \lim_{n \rightarrow \infty} \int_E f_n$
- ”(Leviš ù g  $g_k (U \text{ 6[š ”ü—Fatou}$

### 6.3 Á

šl

$$g_k(x) = \inf_{n \geq k} f_n(x); \quad k = 1; 2; \dots;$$

$$g_k(x) \leq g_{k+1}(x) (U \checkmark)$$

$$\lim_{k \rightarrow \infty} g_k(x) = \lim_{n \rightarrow \infty} f_n(x):$$

$$\int_E g_k(x) dx = \int_E f_n(x) dx; \quad n \geq k;$$

ùúš k ä n ! 1 Ö •P

$$\lim_{k \rightarrow \infty} \int_E g_k(x) dx = \lim_{n \rightarrow \infty} \int_E f_n(x) dx:$$

ù g k ”(LeviU 6[š

$$\lim_{k \rightarrow \infty} \int_E g_k(x) dx = \lim_{n \rightarrow \infty} \int_E f_n(x) dx:$$

ä k ! 1 —

$$\lim_{n \rightarrow \infty} \int_E f_n(x) dx = \lim_{n \rightarrow \infty} \int_E f_n(x) dx:$$

## 7 yī š

### 7.1 š ...<sup>1</sup>

$\frac{3}{4} \text{ ff } f_k(x) g / iK \text{Æ } E \text{ R } ^n \text{ „} iK \acute{y}p$

$$\int_E \bigcup_{k=1}^n f_k(x) dx = \bigcup_{k=1}^n \int_E f_k(x) dx:$$

### 7.2 Á ĭ

- è ĆU ä S  $S_n(x) = \bigcup_{k=1}^n f_k(x) \text{ S } _n(x) \wedge U \check{z} \lim_{n \rightarrow \infty} S_n(x) =$
- "(Leviš ù S  $S_n(x) \not\in (U \text{ š } \acute{O} \text{ P } \text{Æ} \text{ „} i \text{ ' } \text{—} \acute{O}^o$

### 7.3 Á

$$\text{ä S } _n(x) = \bigcup_{k=1}^n f_k(x)$$

$$0 \leq S_1(x) \leq S_2(x) \leq S_n(x) \leq \dots; \lim_{n \rightarrow \infty} S_n(x) = \bigcup_{k=1}^{\infty} f_k(x):$$

1LeviU š

$$\lim_{n \rightarrow \infty} \int_E S_n(x) dx = \int_E \bigcup_{k=1}^{\infty} f_k(x) dx:$$

Èà P\* iKýp Ć „ ĭ IŽ ĭ „ Ć

$$\int_E S_n(x) dx = \bigcup_{k=1}^n \int_E f_k(x) dx;$$

ä n ! 1 —

$$\int_E \bigcup_{k=1}^{\infty} f_k(x) dx = \bigcup_{k=1}^{\infty} \int_E f_k(x) dx:$$



## 8 ÎÔ<(Fubini) X...)(Tonelli)š

### 8.1 š ...<sup>1</sup>

$\frac{3}{4} (X_1;_1) (X_2;_2) : - PK'_1 z \hat{o} f(x; y) / X_1 X_2$  „İKýp 1. X  
...)š (^ Åb) å f 0

$$\begin{aligned} \int_{X_1 \times X_2} f d(\mu_1 \otimes \mu_2) &= \int_{X_1} \int_{X_2} f d\mu_2 d\mu_1 \\ &= \int_{X_2} \int_{X_1} f d\mu_1 d\mu_2: \end{aligned}$$

2. ÎÔ<š (İİÅb) å f (X<sub>1</sub> X<sub>2</sub> İİ İ Ě /İ àN  
P

### 8.2 Á İ

- y•ýp 1XİKİšİ ÓùİİKéb„y•ýp Ě
- €Uýp ž““•0^ €Uýp
- ^ İKýp Leviš “•0^ İKýp(X...))
- İİýp ã:c è “•0 ,İİýp(İÔ<)

### 8.3 Á

#### 8.3.1 y•ýpÅb

$$\frac{3}{4} A = A_1 \otimes A_2 /İKéb f = \int_{X_1 \times X_2} A d(\mu_1 \otimes \mu_2) = (\mu_1 \otimes \mu_2)(A) = \mu_1(A_1) \mu_2(A_2):$$

æ<sup>1b</sup>

$$\begin{aligned} \int_{X_1} \int_{X_2} A d\mu_2 d\mu_1 &= \int_{X_1} A_1(x) \mu_2(A_2) d\mu_1 \\ &= \mu_1(A_1) \mu_2(A_2); \end{aligned}$$

æ /İ \_İžå< Ó° Ě

### 8.3.2 ^ €UýpÅb

$$\frac{3}{4}f = \sum_{i=1}^m c_i A_i (c_i = 0 \text{ A } i \text{ : ikéb } 1 \text{ z"}$$

$$\int_{X_1 \times X_2} f(d_1, d_2) = \int_{X_1} \int_{X_2} f(d_1, d_2) d_1 d_2$$

8.3.3 ^ ĭKýpÅb(X...)š )

$$X(\wedge^k U) \cong \bigoplus_{n \geq k} \frac{f_n d(1, 2)}{n!} = \lim_{n \rightarrow \infty} \frac{f_n d(1, 2)}{n!}.$$

$$\int_{\mathbb{R}^n} f_n d(\mu_1 + \mu_2) = \int_{\mathbb{R}^n} f_n d\mu_1 + \int_{\mathbb{R}^n} f_n d\mu_2$$

$$\lim_{n \rightarrow \infty} \frac{f_n(x_1, x_2)}{f(x_1, x_2)} = 1$$

ĩÁæ /!ĩ øl

### 8.3.4 ģīpÅb(ìÔ<š )

$\ddot{a} f = f^{+} f^{-} (f^{+} - 0 f^{-} - 0 + :c\ddot{e} - \ddot{e}) \ddot{a} f \ddot{u} E f^{+} f^{-} G \ddot{u}$

$$\begin{aligned} Z_{x_1 \times x_2} f^+ d(1, 2) &= \frac{Z}{Z^{x_1}} \frac{Z}{Z^{x_2}} f^+ d_2 d_1 \\ &= \frac{f^+ d_1 d_2}{x_2 x_1}; \\ Z_{x_1 \times x_2} f^- d(1, 2) &= \frac{Z}{Z^{x_1}} \frac{Z}{Z^{x_2}} f^- d_2 d_1 \\ &= \frac{f^- d_1 d_2}{x_2 x_1} \end{aligned}$$

\$ øĩ—

$$\begin{aligned} Z_{X_1 X_2} f d(1, 2) &= \frac{Z_{X_1} Z_{X_2} f d_2 d_1}{Z_{X_2} Z_{X_1}} \\ &= f d_1 d_2: \end{aligned}$$

Èà f ïĩ E/!ĩ „«iýpàN P